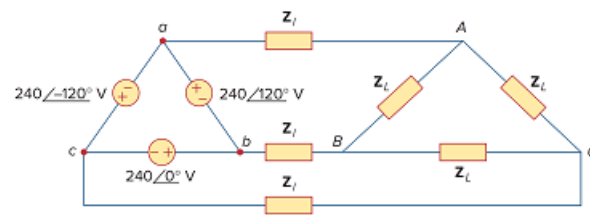


Problem

Find the line currents I_{aA} , I_{bB} , and I_{cC} in the three-phase network of Fig. 12.53. Take $\mathbf{Z}_L = (114 + j87) \Omega$ and $\mathbf{Z}_l = (2 + j) \Omega$.

Figure 12.53



Step-by-step solution

Step 1 of 5

Refer to the Figure 12.53 in the textbook.

Convert the balanced delta load into balanced wye load.

Use the following formula, to convert the load into wye load.

$$Z_{L(Y)} = \frac{Z_{L(\Delta)}}{3}$$

Substitute $(114 + j87) \Omega$ for $Z_{L(\Delta)}$ in the equation.

$$\begin{aligned} Z_{L(Y)} &= \frac{(114 + j87)}{3} \\ &= (38 + j29) \Omega \end{aligned}$$

Step 2 of 5

Now convert the Delta source into Wye-source.

In delta source and wye source, the line voltages are same.

Now, find the phase voltage using following formula:

$$V_{ph} = \frac{V_{line}}{\sqrt{3}}$$

Determine the phase voltages.

$$\begin{aligned} V_1 &= \frac{240 \angle 0}{\sqrt{3}} \angle -30^\circ \\ &= 138.56 \angle -30^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} V_2 &= \frac{240 \angle 120^\circ}{\sqrt{3}} \angle -30^\circ \\ &= 138.56 \angle 90^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} V_3 &= \frac{240 \angle -120^\circ}{\sqrt{3}} \angle -30^\circ \\ &= 138.56 \angle -150^\circ \text{ V} \end{aligned}$$

Step 3 of 5

Therefore, the reduced circuit diagram is,

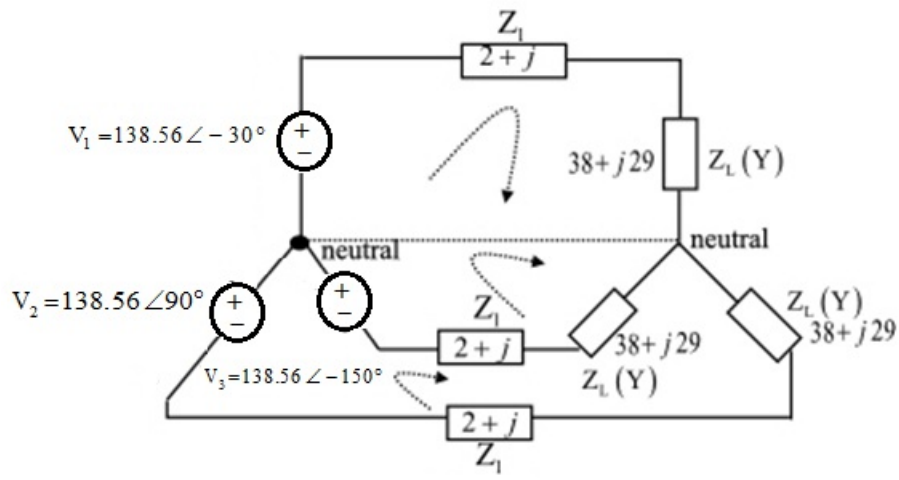


Figure 1

Step 4 of 5

Apply Kirchhoff's voltage law in the loops to find the line currents, I_{aA}, I_{bB}, I_{cC} .

Determine the value of I_{aA} .

$$138.56 \angle -30^\circ = I_{aA} (2 + j + 38 + j29)$$

$$I_{aA} = \frac{138.56 \angle -30^\circ}{40 + j30}$$

$$= 2.77 \angle -66.87^\circ \text{ A}$$

Determine the value of I_{bB} .

$$138.56 \angle -150^\circ = I_{bB} (2 + j + 38 + j29)$$

$$I_{bB} = \frac{138.56 \angle -150^\circ}{40 + j30}$$

$$= 2.77 \angle -186.87^\circ \text{ A}$$

Step 5 of 5

Determine the value of I_{cC} .

$$138.56 \angle 90^\circ = I_{cC} (2 + j + 38 + j29)$$

$$I_{cC} = \frac{138.56 \angle 90^\circ}{40 + j30}$$

$$= 2.77 \angle 53.13^\circ \text{ A}$$

Therefore, the values of line currents I_{aA}, I_{bB} and I_{cC} are as follows:

$$\boxed{2.77 \angle -66.87^\circ \text{ A}}, \boxed{2.77 \angle -186.87^\circ \text{ A}} \text{ and } \boxed{2.77 \angle 53.13^\circ \text{ A}}.$$